

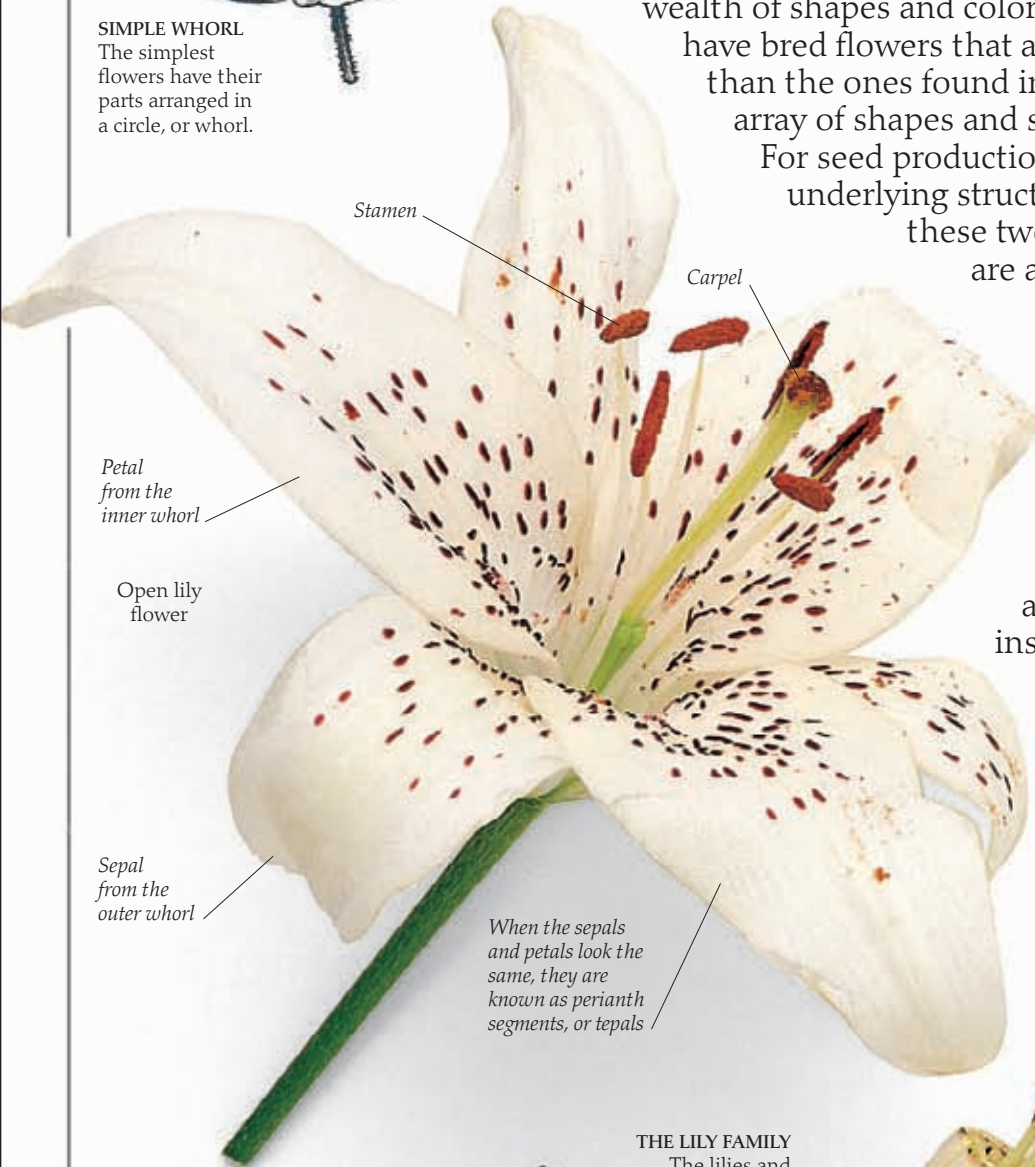
A simple flower dissected



SIMPLE WHORL
The simplest flowers have their parts arranged in a circle, or whorl.

FLOWERS HAVE BECOME extraordinarily varied during the course of evolution. Nature has produced them in a tremendous wealth of shapes and colors. Added to this profusion, people have bred flowers that are even more brilliant or bizarre than the ones found in the wild. But behind this baffling array of shapes and sizes there is a common pattern.

For seed production, all flowers use the same underlying structures. The lily flower shown on these two pages is quite simple—its parts are all separate, and they can all be seen clearly. They fall into three groups. The male parts (the stamens) produce the pollen, while the female parts (the carpel) include the ovary, where seeds are produced. Around both the male and the female parts are sepals and petals, which attract insects to the flower.



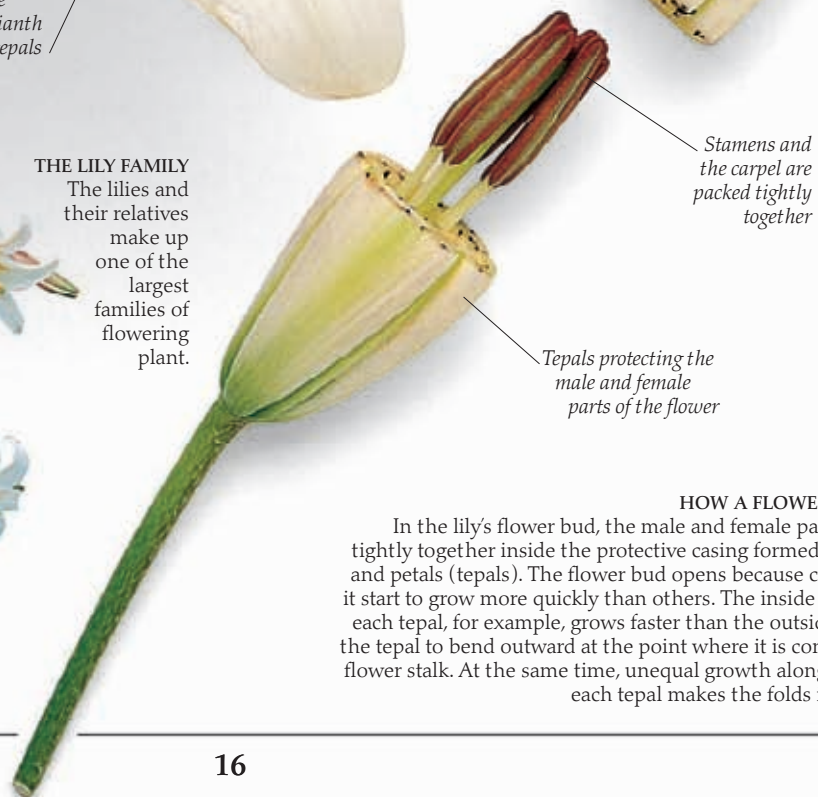
When the sepals and petals look the same, they are known as perianth segments, or tepals



Lily flower bud



THE LILY FAMILY
The lilies and their relatives make up one of the largest families of flowering plant.



HOW A FLOWER BUD OPENS

In the lily's flower bud, the male and female parts are packed tightly together inside the protective casing formed by the sepals and petals (tepals). The flower bud opens because certain parts of it start to grow more quickly than others. The inside of the base of each tepal, for example, grows faster than the outside. This forces the tepal to bend outward at the point where it is connected to the flower stalk. At the same time, unequal growth along the edges of each tepal makes the folds in it open out.